

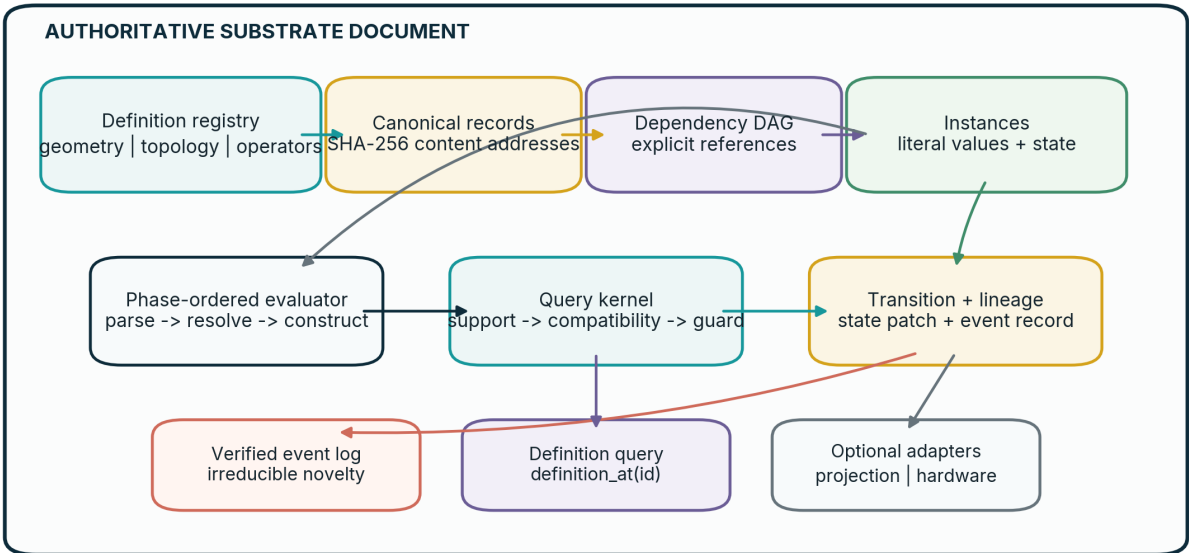
UNIFIED GEOMETRIC-TOPOLOGICAL SUBSTRATE

UGTS-KC 3.6

Literal Referential Definitions, Dutch Phonetic Number Hinges, Geometric Calculus and Topology Patterns

UGTS-KC 3.6: definitions live inside the substrate

A literal, content-addressed definition graph drives instances, queries and events.



Referential closure is bounded: definitions may describe other definitions, but unresolved IDs, invalid hashes and cycles are rejected.

Version

3.6.0, 17 August 2026

Requester attribution

Tom Klootwijk; NL200678942; 10-07-1990. Supplied by the requester and not independently verified.

Technical scope

36 v3.6 operators, 20 pattern recipes, 100-entry Dutch number profile, JSON Schema, literal example substrate, dependency-free Python runtime and 36 passing tests.

Evidence rule

Source-derived and engineering material remain distinct from corrections and rejections.

Abstract

The version 3.6 upgrade changes the substrate from a system that merely *uses* geometric and topological definitions into a system that can *carry, resolve and verify those definitions inside itself*. A definition is now a first-class substrate node with a stable ID, typed domain and codomain, explicit dependencies, evaluation phase, provenance class and canonical SHA-256 content address. Instances reference definitions; pipelines reference operator definitions; queries may inspect both state and the definitions that produced it.

The attached Dutch-number dialogue contributes several reusable structures after unrelated subject matter and unsupported extrapolation are removed: radix carry thresholds, active-bit filtering, a chosen count-to-simplex embedding, Pascal parity, local log-polar addressing, one-bit masks, implicit boundaries and hinge language. The central new inference is a clean Dutch number-order calculus. Decimal place order and Dutch spoken order are represented as separate charts. Regular teens place a unit root before *tien*; non-multiple tens from 21 through 99 use *unit + en + tens*. The conjunction *en* is treated as a connector hinge: it has no decimal place magnitude, yet it is structurally necessary in the spoken graph.

The package implements a bounded 0–99 Dutch profile, compares pronunciation-pulse counts with binary Hamming weight without claiming numeric identity, and embeds any declared pulse count as a point, segment or regular polygon. For 19, the decimal value remains 19; binary 10011 has three active bits; the shipped profile segments *negentien* as *ne | gen | tien*; and three declared pulses may be embedded as a triangle. The equality is explicitly a feature-count coincidence, not a law connecting language, arithmetic and physics.

DECISION

UGTS-KC 3.6 is a literal referential state-and-event substrate. Its authoritative objects are typed definitions, instances, relations, supports, compatibility gates, hinges, events, transitions and lineage. Rendering, physical devices and narrative associations remain optional downstream uses and cannot redefine the mathematical core.

Contents

1	Version 3.6 decision and scope	1
1.1	What changed	1
1.2	Deliverables	1
2	Evidence discipline and source analysis	1
2.1	Admission rule	1
2.2	Hidden tricks extracted from the attached documents	2
3	Literal referential substrate architecture	3
3.1	World object	3
3.2	Definition record	3
3.3	Definition queries	4
4	Geometry definition calculus	4
4.1	Parametric and implicit forms	4
4.2	Radix shells and log-polar charts	4
4.3	Pulse geometry	5
5	Topology and hinge calculus	5
5.1	Topology as explicit gluing	5
5.2	Generic hinge	5
5.3	Composition order	6

6 Dutch phonetic-number calculus	6
6.1 Profile boundary	6
6.2 Place order and spoken order	6
6.3 Syllable-pulse operator	7
6.4 Worked example: 19	7
6.5 Bounded atlas result	8
6.6 Selected lexicon rows	8
7 Kinematic calculus, guards and dynamics	9
7.1 Closed-form motion	9
7.2 Verified event condition	9
7.3 Operation order	9
8 Reusable pattern library	10
9 Literal example substrate	11
9.1 In-substrate definitions	11
9.2 Schema excerpt	11
9.3 Execution result	12
10 Reference implementation and validation	12
10.1 Package modules	12
10.2 Test result	12
10.3 Reproduction commands	13
11 Claims ledger	13
12 Acceptance and kill criteria	14
13 Conclusion	14
A Complete v3.6 operator catalog	15
B Source register and hashes	16
C Package layout	16

1 Version 3.6 decision and scope

1.1 What changed

The earlier supplied reports already establish the mature query-first sequence: finite grammar, typed state, local support, compatibility, relation guard, earliest valid event, transition and lineage. They also insist that one-bit values remain narrow flags and that topology is an explicit routing or gluing rule rather than material magic. UGTS-KC 3.6 retains those boundaries and adds six architectural commitments:

1. **Definitions are substrate data.** Geometry, topology, phonetic profiles, hinges, schedules and operators are stored in the same literal document as instances and pipelines.
2. **References are resolvable.** Every dependency and `definition_ref` must resolve to a unique definition ID.
3. **Definitions are content-addressed.** Canonical JSON records are hashed, excluding the hash field itself.
4. **Operation order is explicit.** A normative ten-phase schedule removes ambiguity about which transformations occur first.
5. **Dutch number morphology is typed.** Numeric place order, orthography, pronunciation-like segments, morphemes, spoken order and hinge nodes are separate fields.
6. **Cross-domain matches remain metadata.** Equal counts may seed patterns, but do not become numeric identity or physical causation.

1.2 Deliverables

The accompanying ZIP contains:

- this PDF and its editable LaTeX source;
- a JSON Schema and a fully literal example substrate with content hashes;
- 36 v3.6 operator definitions and 20 reusable pattern recipes in CSV and JSON;
- a bounded Dutch 0–99 profile in CSV and JSON, plus a pulse-match extract;
- dependency-free Python modules for canonicalization, geometry, topology, Dutch number profiles, definition loading and pipeline execution;
- original diagrams, a reproducible demonstration and 36 passing unit tests;
- privacy-safe source notes, SHA-256 source hashes and package checksums.

BOUNDARY

The v3.6 document is a technical specification and reference implementation. It is not independent scientific validation, a patent grant, proof of authorship priority, a physical-device benchmark or evidence that numerical coincidences establish hidden causal laws.

2 Evidence discipline and source analysis

2.1 Admission rule

A motif enters the technical core only if it can be expressed as at least one of the following: a typed literal, coordinate map, geometric definition, topology/gluing map, hinge, support predicate, compatibility predicate, relation guard, event solver, transition, invariant, lineage rule, bounded grammar, evaluation schedule or measurable error/resource quantity.

The package uses the following dispositions:

RETAIN	Exact or directly reusable operator.
ENGINEERING	New typed implementation or structural inference that preserves a source motif.
CORRECTED	Useful mechanism kept after an explicit mathematical, dimensional or semantic correction.
BOUNDED	Valid only under a declared language profile, schema, relation family, tolerance or finite range.
DEMOTE	Analogy or research prompt, not a core operator.
REJECT	Unsupported totalizing, physical or causal claim.

2.2 Hidden tricks extracted from the attached documents

Motif	Reusable mechanism	Boundary or correction
Radix carry threshold	Digit-count changes at b^k ; use powers of the radix as scale events or shell indices.	The threshold is an exact number-system fact, not proof that physical geometry is inherently binary or decimal.
Active bits and "zero as silence"	Extract set-bit positions and Hamming weight as a sparse feature projection.	Deleting zeros from a displayed feature string never changes the numeric value.
Three active units	Map a declared count of three to three non-collinear vertices.	The triangle is a chosen embedding; three is not automatically a geometric triangle in every domain.
Pascal parity	Use $\binom{n}{k} \bmod 2$ as a deterministic Sierpiński occupancy test.	With zero-based rows, $2^m - 1$ rows are all odd; generic 2^m rows are not.
Combination count	$\binom{5}{3} = 10$ counts placements of three selected positions among five.	The result is not a causal proof of the decimal 9–10 threshold.
Log-polar chart	$\rho = \ln(r/r_0)$ converts radial scaling into additive displacement.	The origin remains singular and requires an explicit core chart.
One-bit jitter	A bit may select a deterministic dither threshold, route, latch or mask.	It redistributes error; it does not guarantee perfect anti-aliasing, infinite resolution or zero cost.
SDF language	Use an implicit field sign and promote $f = 0$ to an event relation.	A generic implicit field is not automatically an exact signed-distance function.
Cut/unroll glyph morph	Edit a boundary graph by cutting an edge, releasing an endpoint and interpolating curve controls.	Intermediate fields may lose exact-distance status; topology state changes at an explicit event.
Parity hinge	A one-bit state selects one of two declared maps.	The bit is not continuous state, geometry, uncertainty or identity.
Double-vacuum motif	Same coordinate, different sheet/phase/orientation may fail compatibility.	Co-location is necessary but not sufficient for coupling.
Möbius/Klein motifs	Implement quotient/gluing maps with explicit orientation and sheet transport.	No physical self-intersection, self-assembly or energy creation follows.
Hourglass/pinch motif	Four finite sectors meet at a guarded routing locus.	The pinch is an event/router, not a destructive singularity.
Self-reference and strange loops	Represent definitions and cycles as a graph with budgets and cycle policy.	Undeclared recursion and unbounded self-generation are rejected.
Dutch number order	Treat decimal place order and spoken morphological order as separate charts connected by a typed hinge map.	This is a language-profile transform, not changed arithmetic.

CORRECTION

The new dialogue contains several persuasive but unsupported escalations: generic SDFs described as exact distances, one-bit jitter described as eliminating aliasing, log-polar coordinates described as infinite storage-free resolution, personal numbers treated as physical invariants, and historical/geological facts inferred from decimal patterns. UGTS-KC 3.6 keeps the reusable operators and rejects those conclusions.

3 Literal referential substrate architecture

3.1 World object

The v3.6 world is defined as

$$\mathcal{W}_{3.6} = (\mathcal{D}, \mathcal{X}, \mathcal{G}, \mathcal{R}, \mathcal{S}, \mathcal{C}, \mathcal{H}, \mathcal{E}, \mathcal{T}, \mathcal{I}, \mathcal{L}, \mathcal{P}),$$

where $\mathcal{D}$ is the in-substrate definition registry, $\mathcal{X}$ instances, $\mathcal{G}$ bounded grammar, $\mathcal{R}$ relations/fields, $\mathcal{S}$ support predicates, $\mathcal{C}$ compatibility, $\mathcal{H}$ hinges, $\mathcal{E}$ events, $\mathcal{T}$ transitions, $\mathcal{I}$ invariants, $\mathcal{L}$ lineage and external novelty log, and $\mathcal{P}$ phase-ordered pipelines.

The architectural change is the explicit $\mathcal{D}$ component. A substrate no longer depends on prose outside the data to explain what sphere, `nl_number_lexeme`, `mobius_quotient` or `guard` means. Those definitions are themselves queryable records.

3.2 Definition record

FORMAL DEFINITION

A literal definition node is

$$d = (id, kind, A, B, \theta, Dep, phase, cap, inv, prov, h),$$

with domain A , codomain B , literal parameters θ , dependency IDs Dep , evaluation phase, capabilities, invariants, provenance and content hash h .

$$h = \text{SHA256}(\text{canonicalJSON}(d \setminus \{h\})).$$

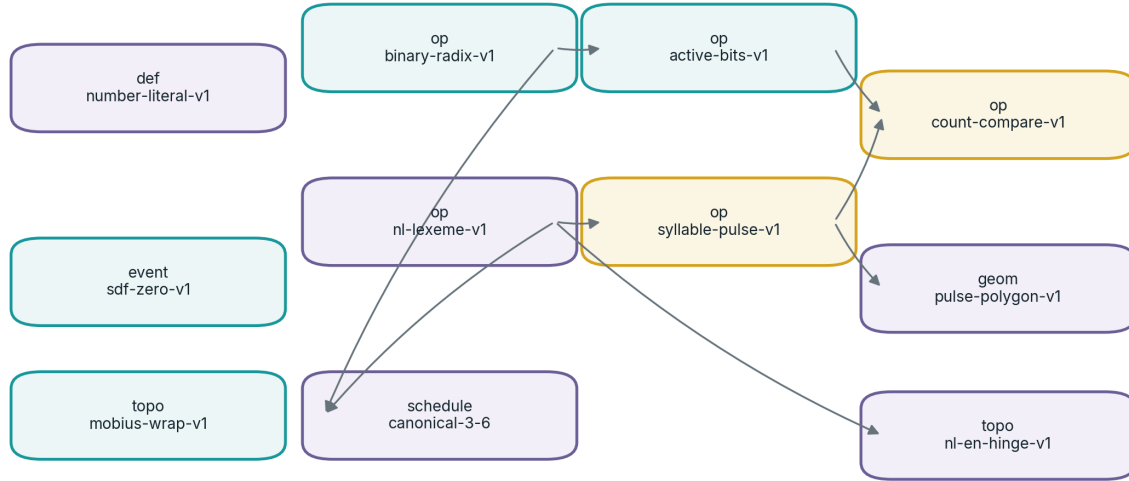
The hash is verification metadata, not a cryptographic ownership claim.

An instance is

$$x = (id, definition_ref, literal, state),$$

and a pipeline is an ordered sequence of definition IDs. Referential closure requires every reference to resolve, every content hash to verify and the dependency graph to admit a topological order.

Example definition dependency graph



Content hashes verify nodes; the loader rejects unknown IDs and dependency cycles.

Figure 1: The shipped example contains literal definition nodes and explicit dependency edges.

3.3 Definition queries

The minimum referential API is:

- `definition_at(id)` – return the exact typed record and content hash;
- `definition_order()` – return a deterministic dependency order;
- `instances_of(definition_id)` – enumerate literal uses;
- `trace(pipeline, instance)` – return ordered definition IDs, operator kinds and typed outputs;
- `verify_definition(id)` – recompute the canonical content hash;
- `explain_reference(id)` – return dependencies, provenance and evaluation phase.

4 Geometry definition calculus

4.1 Parametric and implicit forms

A parametric geometric definition has the form

$$\gamma : U \rightarrow \mathbb{R}^n, \quad u \mapsto \gamma(u),$$

with a declared interval, regularity and frame policy. An implicit definition is

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad f(x) < 0 \text{ inside}, \quad f(x) = 0 \text{ boundary}, \quad f(x) > 0 \text{ outside}.$$

Exact signed-distance status is a capability, not a synonym for “implicit.” CSG sign composition remains useful:

$$f_{A \cup B} = \min(f_A, f_B), \quad f_{A \cap B} = \max(f_A, f_B), \quad f_{A \setminus B} = \max(f_A, -f_B).$$

These formulas preserve inside/outside sign semantics, but smooth blends and generic morphs need not remain exact distances.

4.2 Radix shells and log-polar charts

For radix $b \geq 2$, digit-count transitions occur at b^k . The exact digit count is

$$\text{digits}_b(n) = \begin{cases} 1, & n = 0, \\ \lfloor \log_b n \rfloor + 1, & n > 0. \end{cases}$$

UGTS 3.6 may map those thresholds to local shells $[b^k, b^{k+1})$. This is an indexing choice. The local log-polar chart is

$$\rho = \ln(r/r_0), \quad \theta = \text{atan2}(y, x),$$

with a mandatory `core=true` flag when $r < \varepsilon$.

4.3 Pulse geometry

For a declared pulse sequence of length m , the reference embedding is

$$p_j = r(\cos(\phi + 2\pi j/m), \sin(\phi + 2\pi j/m)), \quad j = 0, \dots, m-1.$$

The special cases are one pulse $\mapsto$ one point, two pulses $\mapsto$ a segment and three pulses $\mapsto$ a triangle. Four or more pulses yield a regular polygon. Pulse identity and source positions remain metadata; geometry never replaces them.

5 Topology and hinge calculus

5.1 Topology as explicit gluing

A quotient/gluing definition names entry and exit ports, coordinate transport, orientation transport, optional sheet change and lineage update. A Möbius-style horizontal wrap can be written

$$(0, y, o) \sim (w, h - y, -o).$$

The explicit state prevents a projected self-intersection from being mistaken for a physical collision.

Topology is explicit gluing, sheet state and routing

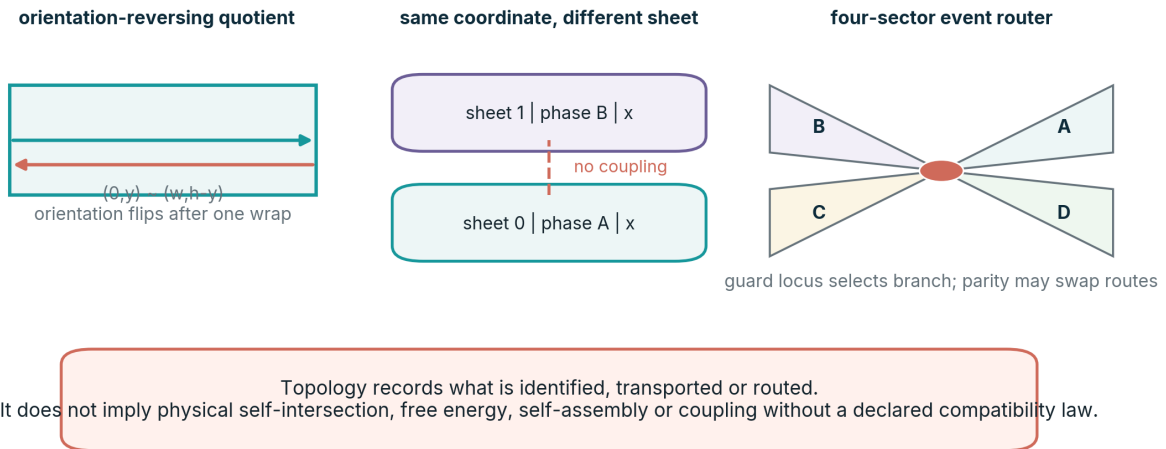


Figure 2: Three distinct topology mechanisms: quotient gluing, compatibility-separated sheets and finite event routing.

5.2 Generic hinge

FORMAL DEFINITION

A hinge is a typed tuple

$$H = (p, h, M, g, F),$$

where p is a pivot or connector locus, h the continuous or discrete hinge state, M_h the selected map, g a guard for state change, and F declared invariants or fixed subspaces.

A continuous planar hinge about pivot p uses

$$x'(t) = p + R(\theta(t))(x - p).$$

Its velocity is

$$\dot{x}'(t) = \dot{\theta}(t)JR(\theta(t))(x - p), \quad J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

A discrete parity hinge selects M_0 or M_1 only after a guarded event. A connector hinge inserts a graph node that may have zero numeric magnitude but nonzero structural role.

5.3 Composition order

Hinge composition is generally non-commutative:

$$H_2 \circ H_1 \neq H_1 \circ H_2.$$

For an articulated chain,

$$T(q) = T_n(q_n) \cdots T_2(q_2)T_1(q_1).$$

UGTS-KC 3.6 stores that order literally. It does not sort transforms by name or silently commute them.

Hinge calculus: one interface, three concrete forms

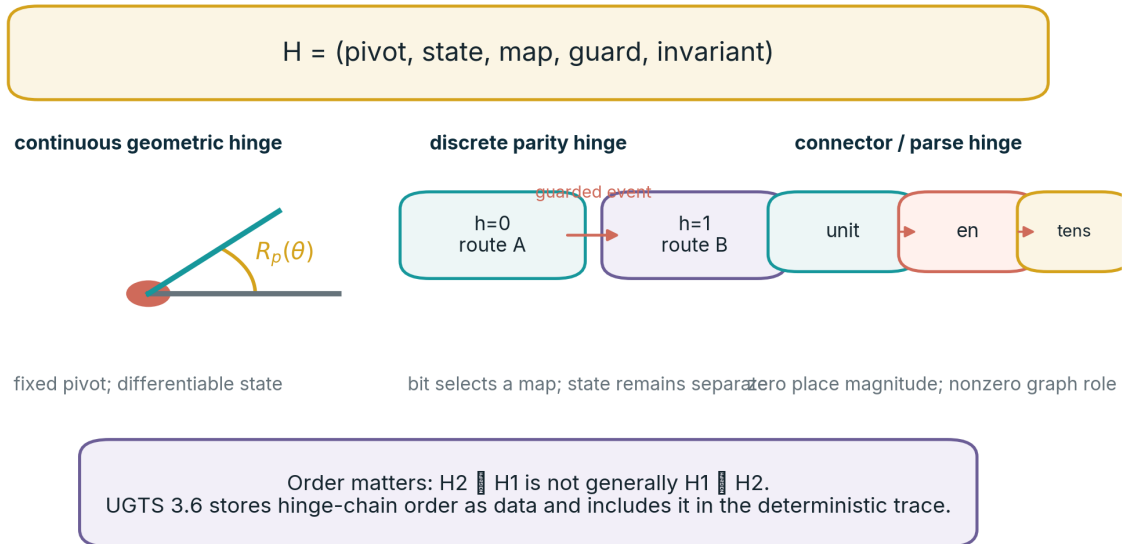


Figure 3: Continuous rotation, a discrete parity selector and a connector hinge share one typed interface.

6 Dutch phonetic-number calculus

6.1 Profile boundary

The shipped `phon:n1-NL-standard-0-99-v1` profile covers integers 0 through 99. It records standard orthography, pronunciation-like segments, morphemes, decimal place order, spoken semantic order, hinge kind, binary spelling and Hamming weight. The segment list is a versioned engineering profile, not an IPA transcription or a claim about every dialect.

6.2 Place order and spoken order

For a non-multiple ten $n = 10t + u$ with $t \in \{2, \dots, 9\}$ and $u \in \{1, \dots, 9\}$, decimal place semantics are

$$P(n) = (10t, u),$$

while the standard Dutch spoken graph is represented as

$$S_{nl}(n) = (u, \text{en}, 10t).$$

The connector hinge is therefore

$$H_{en}(10t, u) = (u, en, 10t).$$

The invariant is numeric value:

$$\text{value}(P(n)) = \text{value}(S_{nl}(n)) = n.$$

The node *en* has no place value, yet its omission would change the declared spoken graph.

For regular teens 13 through 19, place semantics are $(10, u)$ while morphology places the unit root before the suffix *tien*. Eleven and twelve remain explicit irregular lexemes.

Dutch number order as an explicit connector hinge

Example: 23 has place semantics 20 + 3, but the spoken graph is drie - en - twintig.

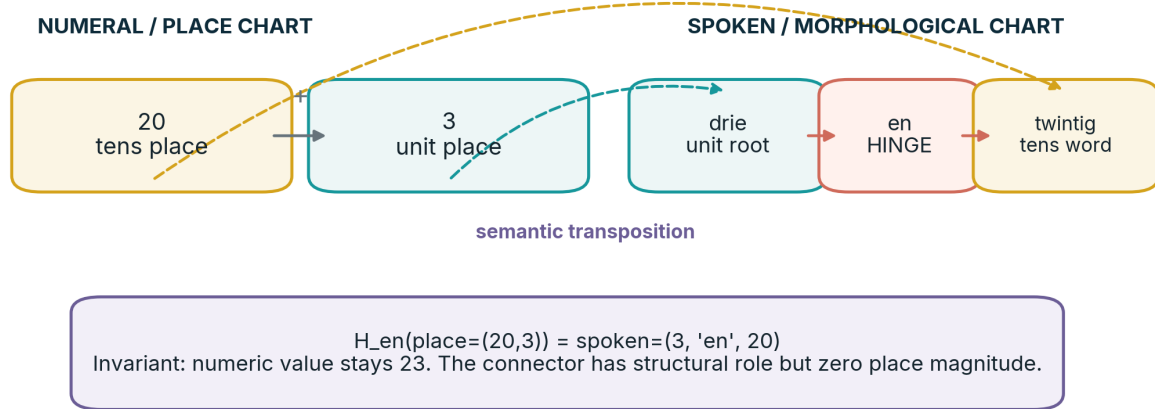


Figure 4: The Dutch *en* connector is modeled as a structural hinge between place and spoken charts.

6.3 Syllable-pulse operator

Let the profile return segments

$$\Sigma_{nl}(n) = (s_1, \dots, s_m).$$

The pulse projection is

$$\Pi_{phon}(n) = (1, \dots, 1) \in \{1\}^m,$$

so $|\Pi_{phon}(n)| = m$. Independently, the binary active set is

$$A_2(n) = \{k \mid \text{bit}_k(n) = 1\}, \quad \text{popcount}(n) = |A_2(n)|.$$

The comparison flag is

$$M(n) = \mathbf{1} [|\Pi_{phon}(n)| = |A_2(n)|].$$

The codomain of M is metadata. It is not a rewrite of n .

6.4 Worked example: 19

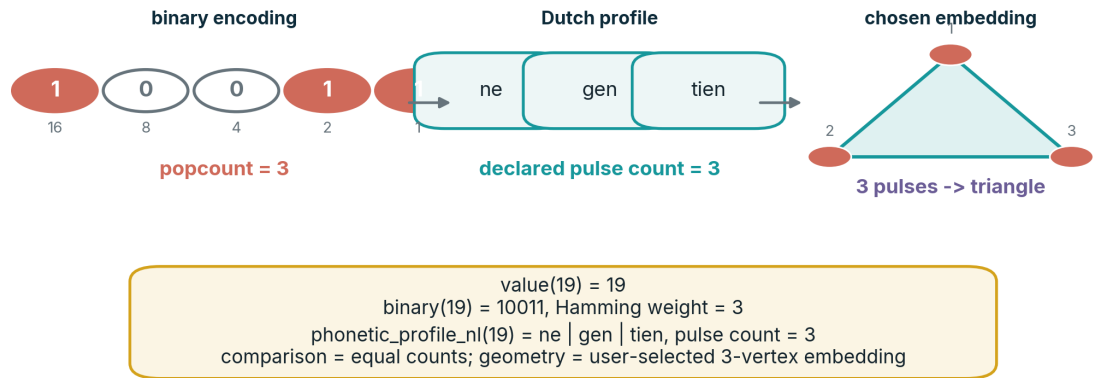
The shipped trace records:

$$\begin{aligned} \text{value} &= 19, \\ \text{binary}_2(19) &= 10011, \\ A_2(19) &= \{0, 1, 4\}, \quad |A_2(19)| = 3, \\ \Sigma_{nl}(19) &= (\text{ne}, \text{gen}, \text{tien}), \quad |\Sigma_{nl}(19)| = 3, \\ M(19) &= 1 \quad \text{with semantics "equal feature counts; no numeric identity."} \end{aligned}$$

A three-vertex pulse embedding yields a triangle. The positions 0, 1 and 4 remain available as source metadata; they are not discarded.

19: three independent representations connected by typed transforms

The shared count is recorded as a coincidence. The decimal value remains 19 throughout.

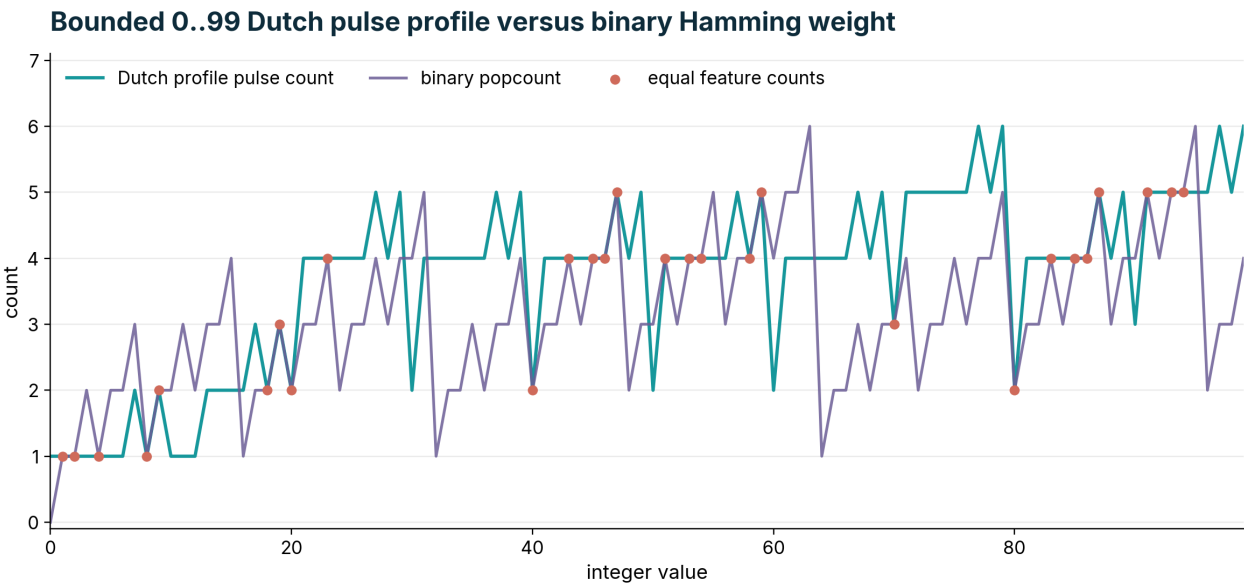


No zero is deleted from the number; zeros are merely inactive positions in one feature projection.

Figure 5: The three-count connection is explicit, typed and bounded.

6.5 Bounded atlas result

Across the shipped 0–99 profile, 28 values have equal pronunciation-pulse count and binary popcount. That result depends on this exact lexicon and segmentation profile. It is useful as a pattern-search dataset, not as a universal law of Dutch or binary arithmetic.



28 matches under the shipped profile; this is a lexicon-dependent comparison, not a numeric law.

Figure 6: Profile pulse counts and binary Hamming weights across 0–99.

6.6 Selected lexicon rows

n	Orthography	Profile segments	Place order	Spoken order	syll.	pop
0	nul	nul	0	0	1	0
1	een	een	1	1	1	1
2	twee	twee	2	2	1	1
7	zeven	ze ven	7	7	2	3

n	Orthography	Profile segments	Place order	Spoken order	syll.	pop
9	negen	ne gen	9	9	2	2
10	tien	tien	10	10	1	2
11	elf	elf	11	11	1	3
12	twalf	twalf	12	12	1	2
13	dertien	der tien	10 3	3 tien	2	3
17	zeventien	ze ven tien	10 7	7 tien	3	2
18	achttien	acht tien	10 8	8 tien	2	2
19	negentien	ne gen tien	10 9	9 tien	3	3
20	twintig	twi gen	20	20	2	2
21	eenentwintig	een en twi gen	20 1	1 en 20	4	3
23	drieëntwintig	drie en twi gen	20 3	3 en 20	4	4
27	zevenentwintig	ze ven en twi gen	20 7	7 en 20	5	4
30	dertig	der tig	30	30	2	4
42	tweeënveertig	twee en veer tig	40 2	2 en 40	4	3
70	zeventig	ze ven tig	70	70	3	3
99	negenennegentig	ne gen en ne gen tig	90 9	9 en 90	6	4

7 Kinematic calculus, guards and dynamics

7.1 Closed-form motion

For constant acceleration,

$$x(t) = p_0 + v_0 \Delta t + \frac{1}{2} a \Delta t^2, \quad v(t) = v_0 + a \Delta t.$$

A geometric relation $f(x) = 0$ becomes a temporal guard

$$g(t) = f(x(t)).$$

Selected line, plane, circle and sphere relations yield linear or quadratic roots for selected trajectories. Generic implicit-field/trajectory pairs require bracketed or certified numerical methods.

7.2 Verified event condition

The next valid event is

$$t^* = \min\{t > t_0 : R_j(q(t)) = 0, S_\alpha(q(t), t) = 1, \chi(q(t), j, t) = 1, \text{conf} \geq \text{conf}_{\min}\}.$$

The event record includes pre-state, post-state, relation residual, crossing direction, solver status, guard margin, confidence, route and lineage.

7.3 Operation order

The normative phases are:

1. parse literals and schemas;
2. normalize units, profiles and sign conventions;
3. resolve and verify definitions;
4. construct geometric and phonetic objects;
5. apply ordered transforms and hinges;
6. test local support;
7. test compatibility;
8. solve and certify relation guards;
9. commit transitions atomically;

10. append lineage and irreducible novelty.

Normative operation order

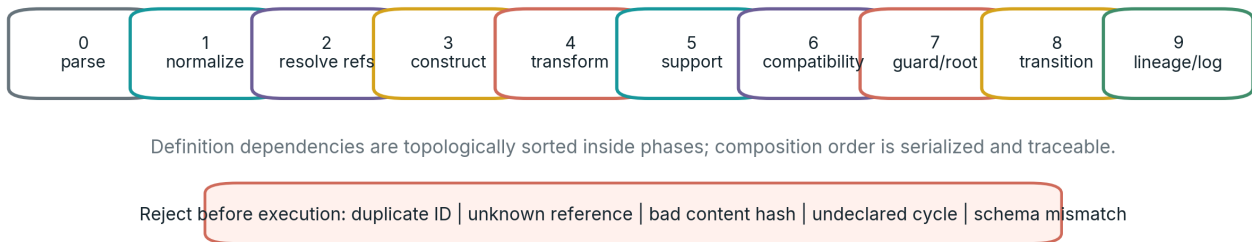


Figure 7: The v3.6 evaluator makes mathematical order part of the substrate rather than hidden implementation detail.

BOUNDARY

A field morph and a topology transition are not the same operation. Continuous interpolation may be evaluated in phase 4, while a topology-class change is committed only in phase 8 after support, compatibility and a verified guard.

8 Reusable pattern library

The 20 recipes below abstract the useful tricks away from the original dialogue subjects. Each recipe names a pipeline and a rule that can be reused for geometry, data structures, animation, procedural construction or event routing.

ID	Pattern	Pipeline	Rule
P36-01	Radix threshold ladder	number -> thresholds	Use b^k boundaries as discrete scale events or shell indices.
P36-02	Active-bit pulse simplex	integer -> set-bit pulses -> polygon	Embed sparse active positions as a chosen simplex or regular polygon while retaining bit positions as metadata.
P36-03	Pascal parity mask	$(n,k) \rightarrow C(n,k) \bmod 2$	Generate a Sierpinski occupancy mask with exact parity tests; do not claim generic power-of-two rows are all odd.
P36-04	Dutch teen suffix hinge	13..19 -> unit-root + tien	Represent morphological order separately from decimal place order.
P36-05	Dutch en-order hinge	21..99 non-multiple tens -> unit + en + tens	Insert a connector node and perform a two-component semantic transposition.
P36-06	Cut-open boundary graph	closed edge -> cut port pair	Change a loop constraint into an open path with explicit endpoint identity.
P36-07	Loop-to-leg morph	closed arc -> diagonal segment	Interpolate curve controls while separately recording the topology event.
P36-08	Split-oval antisymmetry	single loop -> two branches	Use a fixed center node and opposite signed displacement fields.
P36-09	Overlap-lens domain	$A, B \rightarrow A \text{ intersection } B$	Treat overlap as geometry, tolerance, blend or join domain only after semantics are declared.
P36-10	SDF-zero event	$f(x(t)) \rightarrow \text{root} \rightarrow \text{transition}$	Promote a boundary crossing from sampling detail to event semantics.
P36-11	Cone support gate	position,axis,radius,angle -> admit/reject	Prune by radial and angular support before relation solving.
P36-12	Nested-shell relevance	radius -> interval index	Use concentric intervals as local multiscale support, not global identity.

ID	Pattern	Pipeline	Rule
P36-13	Möbius orientation wrap	boundary crossing -> coordinate wrap + orientation flip	Implement non-orientability as an explicit quotient map.
P36-14	Klein sheet wrap	boundary crossing -> wrap + orientation and sheet update	Keep projected self-intersection distinct from state coupling.
P36-15	Hourglass route table	guard locus + quadrant + parity -> branch	Route through a finite four-sector table.
P36-16	Log-polar bitmask	Cartesian point -> core flag, rho, theta -> sector bit	Use a compact local admission mask with explicit dimensions and singular core.
P36-17	Golden-angle schedule	index -> low-regularity angular sample	Use as a sample schedule only; no zero-aliasing or zero-cost claim.
P36-18	Bounded grammar recursion	seed + productions + budgets -> definition graph	Stop on depth, symbol, expression or cycle limits.
P36-19	Event-latched field morph	continuous alpha + discrete topology state	Keep visual interpolation and committed topology state separate.
P36-20	Content-addressed definition graph	canonical records -> hashes -> dependency DAG	Make the substrate carry and verify its own operator definitions.

9 Literal example substrate

9.1 In-substrate definitions

The example JSON includes definitions for a literal integer, binary radix encoding, active-bit filtering, a Dutch number lexeme, syllable pulses, count comparison, pulse polygon geometry, a Dutch connector hinge, a Möbius quotient, an SDF-zero event relation and the canonical evaluation schedule. Each record carries a content hash.

FORMAL DEFINITION

The example is not pseudocode. It is accepted by the shipped JSON Schema, loaded by the reference runtime, topologically ordered, hash-verified and executed by the test suite.

9.2 Schema excerpt

```

1 {
2   "$schema": "ugts_kc_3.6.schema.json",
3   "schema_version": "3.6.0",
4   "substrate_id": "ugts:kc:3.6:example:nl-number-hinge",
5   "metadata": {
6     "title": "UGTS-KC 3.6 literal Dutch number-hinge example",
7     "description": "Definitions, instances and execution order are stored in the same substrate document and referenced by stable
8     ↪ IDs.",
9     "evidence_boundary": "Phonetic and binary counts may coincide but are not treated as numeric identity. Topology is explicit
10    ↪ routing/gluing, not physical magic.",
11     "requester_attribution": {
12       "name": "Tom Klootwijk",
13       "identifier": "NL200678942",
14       "date_of_birth": "10-07-1990",
15       "status": "requester-supplied-unverified"
16     }
17   },
18   "definitions": [
19     {
20       "id": "def:number-literal-v1",
21       "kind": "literal_integer",
22       "domain": "{}",
23       "codomain": "integer[0,99]",
24       "evaluation_phase": 0,
25       "dependencies": [],
26       "parameters": {},
27       "provenance": {
28         "class": "engineering-derived",
29         "note": "Literal numeric instance family.",
30         "source_refs": []
31       },
32       "content_hash": "sha256:79ad07905ca506b17b0f5947948a25495bbe3a2386e110b6e4b65226c6f39e16"
33     },
34     {
35       "id": "op:binary-radix-v1",

```

```

34     "kind": "radix_encode",
35     "domain": "integer[0,99]",
36     "codomain": "radix_word",
37     "evaluation_phase": 1,
38     "dependencies": [],
39     "parameters": {
40       "base": 2
41     },
42     "provenance": {
43       "class": "source-derived",
44       "note": "Exact positional encoding.",
45       "source_refs": [
46         "SRC-MD lines 29-40",
47         "UGTS-GN M001-M004"
48       ]
49     },
50     "content_hash": "sha256:35d697e690e3e3eb2f5f8ebd7c343ad73e1cb4c1b95ccd9d76507bbef47bcc50"
51   },
52   {
53     "id": "op:active-bits-v1",
54     "kind": "active_bit_filter",
55     "domain": "integer[0,99]",
56     "codomain": "active_bit_set",
57     "evaluation_phase": 2,
58     "dependencies": [

```

9.3 Execution result

For number:19, the trace ends with a three-vertex polygon and a feature comparison carrying the exact semantics *feature-count coincidence; no numeric identity*. For number:23, it records spoken order [3, "en", 20] and a four-pulse polygon.

10 Reference implementation and validation

10.1 Package modules

src/ugts36/canonical.py	Canonical JSON, SHA-256 content addresses and hash verification.
src/ugts36/phonetics_nl.py	Bounded Dutch 0-99 lexicon, hinges, pulse counts and feature matches.
src/ugts36/geometry.py	Radix operations, active bits, regular embeddings, affine hinges, log-polar chart, implicit fields and kinematics.
src/ugts36/topology.py	Dependency sorting, permutation hinges, Möbius/Klein maps, guard crossing and hourglass routing.
src/ugts36/model.py	Typed definition nodes, substrate loading, reference validation and dependency order.
src/ugts36/runtime.py	Minimal literal pipeline evaluator and deterministic trace.
tests/test_ugts36.py	Thirty-six executable tests.

10.2 Test result

```

test_01_canonical_key_order (test_ugts36.CanonicalTests.test_01_canonical_key_order) ... ok
test_02_hash_is_stable (test_ugts36.CanonicalTests.test_02_hash_is_stable) ... ok
test_03_hash_changes_with_semantics (test_ugts36.CanonicalTests.test_03_hash_changes_with_semantics) ... ok
test_04_attach_and_verify (test_ugts36.CanonicalTests.test_04_attach_and_verify) ... ok
test_13_negentien_profile (test_ugts36.DutchPhoneticTests.test_13_negentien_profile) ... ok
test_14_drieentwintig_profile (test_ugts36.DutchPhoneticTests.test_14_drieentwintig_profile) ... ok
test_15_place_and_spoken_order_are_distinct (test_ugts36.DutchPhoneticTests.test_15_place_and_spoken_order_are_distinct) ... ok
test_16_multiple_of_ten_has_no_connector (test_ugts36.DutchPhoneticTests.test_16_multiple_of_ten_has_no_connector) ... ok
test_17_zeventien_segments (test_ugts36.DutchPhoneticTests.test_17_zeventien_segments) ... ok
test_18_elf_is_irregular (test_ugts36.DutchPhoneticTests.test_18_elf_is_irregular) ... ok
test_19_ninety_nine_profile (test_ugts36.DutchPhoneticTests.test_19_ninety_nine_profile) ... ok
test_20_lexicon_is_complete_and_unique (test_ugts36.DutchPhoneticTests.test_20_lexicon_is_complete_and_unique) ... ok
test_21_pulse_match_set_is_declared (test_ugts36.DutchPhoneticTests.test_21_pulse_match_set_is_declared) ... ok
test_22_profile_bounds (test_ugts36.DutchPhoneticTests.test_22_profile_bounds) ... ok
test_05_digit_count (test_ugts36.GeometryTests.test_05_digit_count) ... ok
test_06_radix_digits (test_ugts36.GeometryTests.test_06_radix_digits) ... ok
test_07_active_bits (test_ugts36.GeometryTests.test_07_active_bits) ... ok
test_08_regular_triangle_has_area (test_ugts36.GeometryTests.test_08_regular_triangle_has_area) ... ok
test_09_point_and_segment_embeddings (test_ugts36.GeometryTests.test_09_point_and_segment_embeddings) ... ok
test_10_rotation_preserves_pivot (test_ugts36.GeometryTests.test_10_rotation_preserves_pivot) ... ok
test_11_logpolar_roundtrip (test_ugts36.GeometryTests.test_11_logpolar_roundtrip) ... ok
test_12_sdf_and_csg_signs (test_ugts36.GeometryTests.test_12_sdf_and_csg_signs) ... ok
test_31_json_schema_validation (test_ugts36.SubstrateRuntimeTests.test_31_json_schema_validation) ... ok
test_32_substrate_load_and_order (test_ugts36.SubstrateRuntimeTests.test_32_substrate_load_and_order) ... ok
test_33_all_definition_hashes_verify (test_ugts36.SubstrateRuntimeTests.test_33_all_definition_hashes_verify) ... ok
test_34_runtime_trace_for_19 (test_ugts36.SubstrateRuntimeTests.test_34_runtime_trace_for_19) ... ok

```

```

test_35_runtime_trace_for_23 (test_ugts36.SubstrateRuntimeTests.test_35_runtime_trace_for_23) ... ok
test_36_missing_reference_is_rejected (test_ugts36.SubstrateRuntimeTests.test_36_missing_reference_is_rejected) ... ok
test_23_permutation_connector_hinge (test_ugts36.TopologyHingeTests.test_23_permutation_connector_hinge) ... ok
test_24_bad_permutation_is_rejected (test_ugts36.TopologyHingeTests.test_24_bad_permutation_is_rejected) ... ok
test_25_affine_composition_is_noncommutative (test_ugts36.TopologyHingeTests.test_25_affine_composition_is_noncommutative) ... ok
test_26_mobius_flip (test_ugts36.TopologyHingeTests.test_26_mobius_flip) ... ok
test_27_klein_flip_and_sheet_toggle (test_ugts36.TopologyHingeTests.test_27_klein_flip_and_sheet_toggle) ... ok
test_28_linear_crossing_time (test_ugts36.TopologyHingeTests.test_28_linear_crossing_time) ... ok
test_29_hourglass_routes (test_ugts36.TopologyHingeTests.test_29_hourglass_routes) ... ok
test_30_topological_order_and_cycle (test_ugts36.TopologyHingeTests.test_30_topological_order_and_cycle) ... ok

```

```

-----
Ran 36 tests in 0.007s

```

```

OK

```

The tests establish internal consistency of the bounded package. They verify canonical hashing, radix transitions, active bits, geometry embeddings, Dutch lexicon classes, the *en* hinge, composition order, quotient maps, guard crossing, dependency cycles, JSON Schema validity and literal pipeline traces. They do not establish physical performance or universal linguistic laws.

10.3 Reproduction commands

```

cd UGTS_KC_3_6_Tom_Klootwijk
PYTHONPATH=src python -m unittest discover -s tests -v
PYTHONPATH=src python examples/demo.py

```

11 Claims ledger

Claim or motif	Disposition	Technical treatment in 3.6
Digit thresholds occur at radix powers.	SUPPORTED	Exact positional-number result; admitted as a shell/index event.
19 has binary Hamming weight three.	SUPPORTED	$19 = 16 + 2 + 1$ and 10011 has three set bits.
The shipped Dutch profile gives three pulses for <i>negentien</i> .	BOUNDED	True under the explicit profile; dialect/tokenizer semantics are versioned.
Therefore 19 equals 111.	REJECT	The active-bit projection has three ones; the numeric binary representation remains 10011.
Three pulses can form a triangle.	ENGINEERING	A declared three-vertex embedding is valid; it is a chosen map.
Pascal parity produces a Sierpiński pattern.	SUPPORTED	Exact parity construction, with corrected row-index statement.
$\binom{5}{3} = 10$ proves the decimal threshold.	DEMOTE	Correct combinatorial count, but no causal proof.
A log-polar LUT removes the origin singularity.	CORRECTED	Scaling becomes translation in ρ ; an explicit core remains.
One-bit jitter eliminates aliasing.	REJECT	It may dither or shape error; quality and cost remain measurable.
A generic SDF gives exact distance everywhere.	CORRECTED	Only valid primitives/transforms claim exact-distance capability.
A one-bit hinge contains complete state.	REJECT	A bit selects route/map/validity; continuous state and lineage remain separate.
Topology creates energy or physical self-assembly.	REJECT	Topology is an explicit gluing/routing abstraction unless an independent physical model is supplied.
Personal dates or historical numbers are geometric invariants.	REJECT	They may be literal metadata, never promoted to causal geometry without an external model and evidence.
Definitions can refer to definitions.	SUPPORTED WITH BOUNDS	References must resolve, hashes verify and dependencies remain acyclic under v3.6.

Claim or motif	Disposition	Technical treatment in 3.6
Dutch <i>en</i> acts as a structural hinge.	ENGINEERING	The connector is a typed graph node between unit and tens in the shipped spoken-order chart.

12 Acceptance and kill criteria

UGTS-KC 3.6 should be rejected or simplified for a target system when any of the following occurs:

- definitions cannot be normalized to typed domains/codomains or stable content addresses;
- unresolved references, ambiguous language profiles or cycles dominate the graph;
- operation order changes results but is not recorded;
- support and compatibility fail to prune enough candidate relations;
- numerical error exceeds the event guard margin or changes event order;
- morphology or phonetic segmentation is treated as universal rather than profile-bound;
- a feature coincidence is used as a causal or physical conclusion;
- event/branch density costs more than the materialization avoided;
- a conventional data model is simpler, faster or more accurate at the same error budget.

13 Conclusion

The useful thread in the attached documents is not a hidden universal code. It is a family of practical transformations: radix thresholds, sparse active positions, parity patterns, boundary edits, local coordinate charts, implicit relations, guarded topology changes, explicit hinges and lineage-aware routing. Version 3.6 packages those transformations into the substrate itself. Every definition has a name, type, dependency set, phase, provenance and content hash; every instance references a definition; every pipeline records its order.

Dutch number language supplies a particularly clean structural example. The numeral 23 stores tens before units, while the spoken form *drieëntwintig* places the unit before the tens word and joins them through *en*. UGTS-KC 3.6 represents that difference as a chart map with a connector hinge. The same discipline keeps the 19/three-pulse observation useful without confusing a memorable pattern with arithmetic identity.

DECISION

Version 3.6 is ready as a bounded, inspectable and executable definition substrate. Its strongest upgrade is referential: the geometry, topology, hinge rules, language profile and operation order are no longer external explanations. They are literal, verifiable objects inside the substrate.

A Complete v3.6 operator catalog

The 36 mechanisms below are the version 3.6 delta namespace. They do not silently renumber the 197-mechanism baseline documented by the supplied GPU-native addendum.

ID	Domain	Mechanism	Normalized technical definition	Disposition / validation
K36-01	Referential core	Literal definition node	A geometry, topology, operator, profile or schedule is stored as a typed definition record inside the substrate.	RETAIN / Schema + tests
K36-02	Referential core	Content-addressed definition	Canonical UTF-8 JSON, excluding the hash field itself, is hashed with SHA-256 to make definition identity verifiable.	ENGINEERING / Implemented
K36-03	Referential core	Explicit dependency edge	Every composed definition lists definition IDs it depends on; hidden imports and implicit global state are forbidden.	RETAIN / Implemented
K36-04	Referential core	Versioned namespace	Definition IDs include stable namespaces while schema_version and profile_version separate compatible from incompatible semantics.	RETAIN / Schema
K36-05	Referential core	Definition-instance separation	A definition describes an operator or shape family; an instance supplies literal parameters and state through definition_ref.	RETAIN / Schema + runtime
K36-06	Referential core	Acyclic resolution rule	Definition dependency graphs must be acyclic unless a future fixed-point kind declares bounds, tolerance and termination.	BOUNDED / Cycle rejection tested
K36-07	Geometry	Typed implicit-field sign	An implicit field declares domain, codomain and sign convention: negative interior, zero boundary, positive exterior.	RETAIN / Specified
K36-08	Geometry	Exact-SDF capability flag	Exact signed-distance status is a capability that only valid primitives and distance-preserving transforms may claim.	CORRECTED / Schema guard
K36-09	Geometry	Parametric curve and sweep	A curve $\gamma(u)$ and a frame or transport rule may generate a swept surface; parameter interval and regularity are explicit.	ENGINEERING / Specified
K36-10	Geometry	Pulse polygon embedding	m declared pulses map to a point, segment or regular m -gon; for $m=3$ the result is a triangle.	ENGINEERING / Implemented
K36-11	Geometry	Radix shell chart	Digit thresholds b^k form exact scale boundaries and may index nested shells without implying that space is inherently radix-based.	RETAIN / Implemented
K36-12	Geometry	Log-polar core chart	$\rho=\ln(r/r_0)$, $\theta=\text{atan2}(y,x)$ is local and carries an explicit core/singularity flag.	RETAIN / Implemented
K36-13	Topology	Explicit quotient/gluing map	Topology is represented by boundary ports, coordinate maps, orientation transport and optional sheet changes.	RETAIN / Implemented examples
K36-14	Topology	Sheet-aware co-location	Equal coordinates do not imply equality or coupling; sheet, phase, orientation, branch and policy remain typed state.	RETAIN / Schema
K36-15	Topology	Orientation transport	Orientation is propagated independently and flips only at declared maps or transitions.	RETAIN / Mobius/Klein tests
K36-16	Topology	Port and branch routing	A guarded crossing selects an explicit output port and branch; the routing table is finite and inspectable.	RETAIN / Specified
K36-17	Topology	Split-merge lineage	Splits create child addresses and merges record all parents; coordinates never replace durable identity.	RETAIN / Schema
K36-18	Topology	Guarded topology change	A morph may vary continuously, but a topology-class change commits only at an event with a declared guard and transition.	CORRECTED / Specified
K36-19	Hinge calculus	Generic hinge tuple	A hinge is $H=(\text{pivot}, \text{state}, \text{map}, \text{guard}, \text{invariant})$ with typed continuous or discrete state.	ENGINEERING / Specified
K36-20	Hinge calculus	Continuous planar hinge	A continuous hinge uses an affine rotation about a fixed pivot; velocity follows the derivative of the transform.	RETAIN / Implemented
K36-21	Hinge calculus	Discrete parity hinge	A one-bit state selects one of two declared maps or routes and does not carry full geometry or identity.	RETAIN / Implemented
K36-22	Hinge calculus	Connector hinge	A zero-magnitude connector can be topologically necessary in an ordered parse graph, as with Dutch en.	ENGINEERING / Lexicon-tested
K36-23	Hinge calculus	Non-commutative hinge chain	Composition order is explicit because $H_2 \circ H_1$ generally differs from $H_1 \circ H_2$.	RETAIN / Tested
K36-24	Hinge calculus	Invariant subspace contract	A hinge may declare fixed points, axes or quantities; the runtime validates them instead of inferring invariance from a name.	ENGINEERING / Specified
K36-25	Dutch phonetics	Bounded Dutch number profile	A versioned lexicon covers standard 0..99 orthography, pronunciation-like segments, morphemes and hinge metadata.	BOUNDED / 100 entries generated
K36-26	Dutch phonetics	Syllable-pulse operator	Each declared pronunciation segment emits one pulse; dialect and tokenizer are profile metadata.	ENGINEERING / Implemented
K36-27	Dutch phonetics	Teen suffix hinge	For regular 13..19 forms, the unit root precedes tien although decimal place semantics are ten plus unit.	RETAIN / Lexicon-tested
K36-28	Dutch phonetics	En connector hinge	For non-multiple tens 21..99, spoken morphology is unit-en-tens while numeral place order is tens-unit.	RETAIN / Lexicon-tested
K36-29	Dutch phonetics	Spoken-place permutation	The place tuple and spoken tuple are separate charts connected by an explicit permutation plus optional connector node.	ENGINEERING / Implemented
K36-30	Dutch phonetics	Feature-count comparison	Binary popcount and phonetic pulse count may be compared; equality is tagged as a coincidence, never numeric identity.	CORRECTED / Implemented
K36-31	Evaluation	Phase-ordered evaluator	The canonical order is parse, normalize, resolve, construct, transform, support, compatibility, guard, transition, lineage.	ENGINEERING / Schedule shipped

ID	Domain	Mechanism	Normalized technical definition	Disposition / validation
K36-32	Evaluation	Support-compatibility-guard discipline	Cheap local support precedes compatibility; event solving occurs only after both gates admit a candidate.	RETAIN / Runtime rule
K36-33	Evaluation	Deterministic trace	Each pipeline step records definition ID, operator kind and typed output to make order and provenance inspectable.	RETAIN / Runtime + demo
K36-34	Evaluation	Evidence disposition	Every source-derived or newly inferred mechanism is labeled RETAIN, ENGINEERING, CORRECTED, BOUNDED, DEMOTE or REJECT.	RETAIN / Catalog
K36-35	Evaluation	Unknown-reference and cycle rejection	Loading fails on missing definition IDs, duplicate IDs, invalid hashes or cyclic dependencies.	RETAIN / Tests
K36-36	Evaluation	Package verification contract	Schema validation, executable tests, source hashes, file hashes and a reproducible demo accompany the PDF.	RETAIN / Delivered

B Source register and hashes

ID	Source	SHA-256 and role
SRC-A	Dutch-number dialogue Markdown	7aa55d4f03e07c6ec058b1dd05554c03730d0f6263803b9364fe34ccce6c1851. Source for phonetic, radix, active-bit, triangle, fractal, log-polar, hinge and claims-boundary analysis.
SRC-B	Unified Geometric-Topological Substrate (UGTS-0)	c991aba24cc939ba680697b62b9e19545131cf34060793490bf149c6e2ab522e. Primary query-first architecture, relation/event calculus, topology and evidence boundary.
SRC-C	UGTS-GN 1.1 GPU-native addendum	3a6b6c57118ca4e2d7d7e577508c3477a095d133791f3708cb56a054285225d2. Normalized catalog, narrow one-bit semantics, precision, ABI, validation and claims ledger.

The raw uploaded sources are not duplicated in the package. The ZIP contains hashes and normalized source notes so that technical traceability is preserved without reproducing unrelated personal or narrative material.

C Package layout

```

UGTS_KC_3_6_Tom_Klootwijk/
README.md
CHANGELOG.md
NOTICE.md
VERSION
pyproject.toml
validation_report.json
report/UGTS_KC_3_6.pdf
report/UGTS_KC_3_6.tex
report/figures/*.png
spec/ugts_kc_3_6.schema.json
spec/operator_catalog_3_6.{csv,json}
spec/pattern_recipes_3_6.{csv,json}
data/dutch_numbers_0_99.{csv,json}
data/dutch_binary_pulse_matches_0_99.csv
examples/ugts_kc_3_6_example.json
examples/demo.py
examples/demo_output.json
src/ugts36/*.py
tests/test_ugts36.py
test_results.txt
sources/source_register.json
sources/source_analysis.md
checksums/SHA256SUMS.txt

```